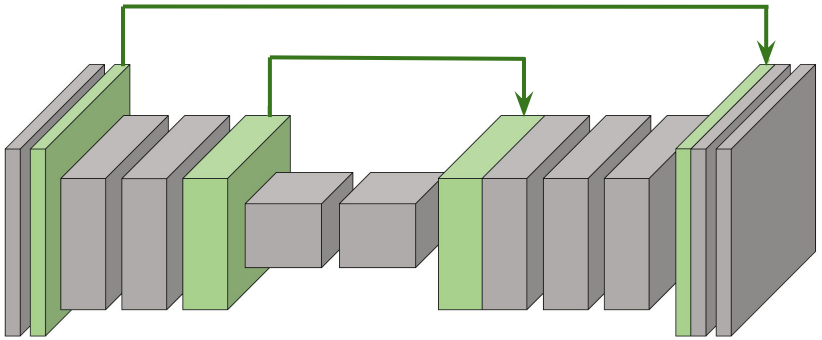
Training of Image2Image translation

Input comes in (X,Y) pairs
Loss: squared loss or absolute loss

$$\mathcal{L}(G(X), Y) = |G(X) - Y|$$
$$\mathcal{L}(G(X), Y) = \|G(X) - Y\|^2$$

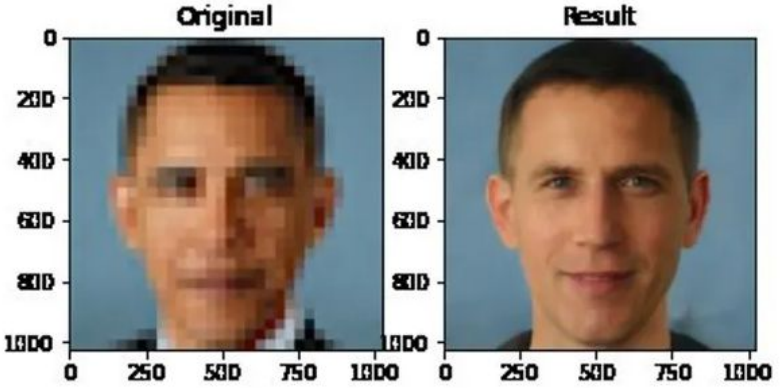
X



Y

Examples

But: Super-resolution inherits biases of training data. Filled in details are not real.



https://www.theverge.com/21298762/face-depixelizer-ai-machine-learning-tool-pulse-stylegan-obama-bias

What happens if we don’t have paired data?

Image Transformations with Deep Learning: Four Key Cases

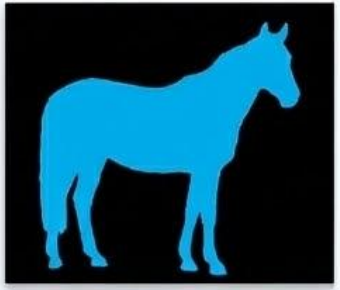
1. Image Classification

→ Output Label: “Horse”

(Input: Image → Output: Label)

2. Paired Image Translation

→



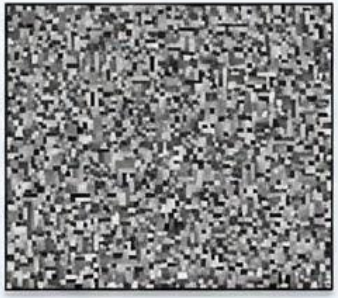
(Input: Image → Output: Image)

3. Unpaired Image Translation

→

(Input: Image → Output: Image)

4. Image Generation

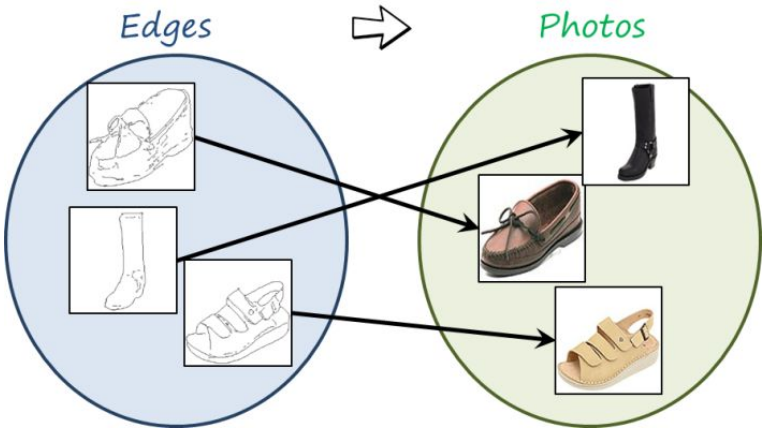


→

(Input: Noise → Output: Image)

Unpaired Image Translation. We have X but no Y!

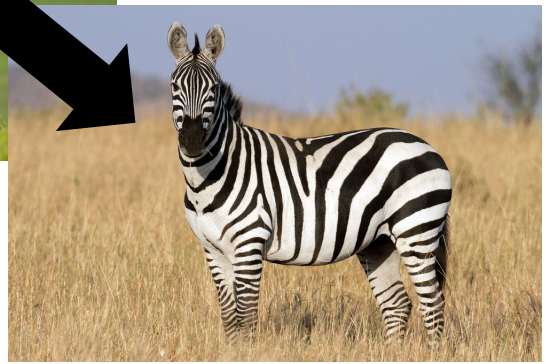
Paired



Generative adversarial networks and image-to-image translation | Luis Herranz

Unpaired

12 Astonishing Facts About Horses

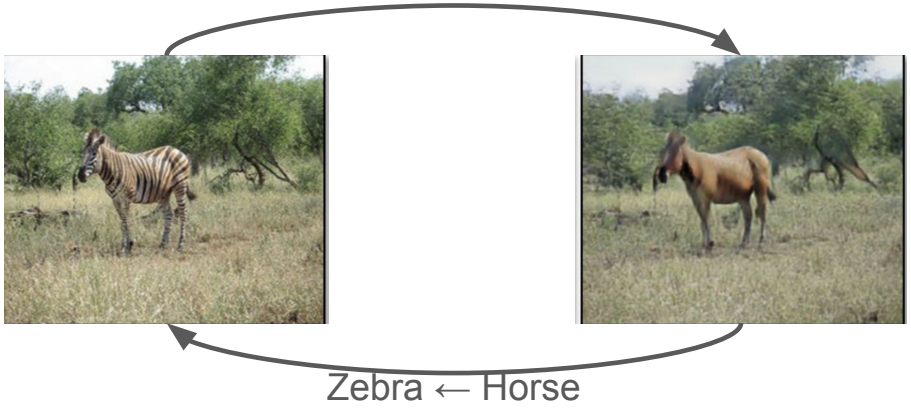


Zebra Facts | Live Science

Paired limitations - it is hard to find exact pairings
We have images of horses, we have images of zebras, but we don't have paired images of what a specific horse would look like as a zebra.

Key Idea: Cycle Consistency

- Image translation should be **invertible**!
  - Translating a zebra to a horse and then back to a zebra should recover the original image
- Cycle consistency!

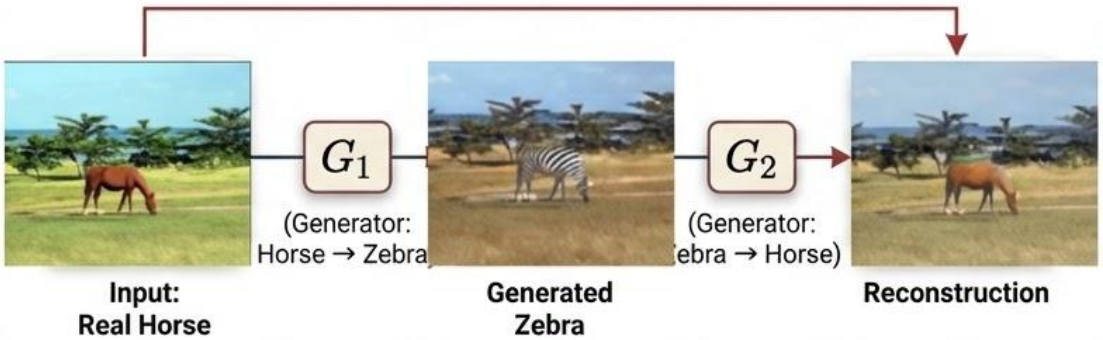


CycleGAN

Cycle Consistency & Reconstruction

$$\|\mathbf{x}_{\text{horse}} - G_2(G_1(\mathbf{x}_{\text{horse}}))\|_2^2$$

Reconstruction Loss: Squared error between original and reconstructed image, enforcing cycle consistency.



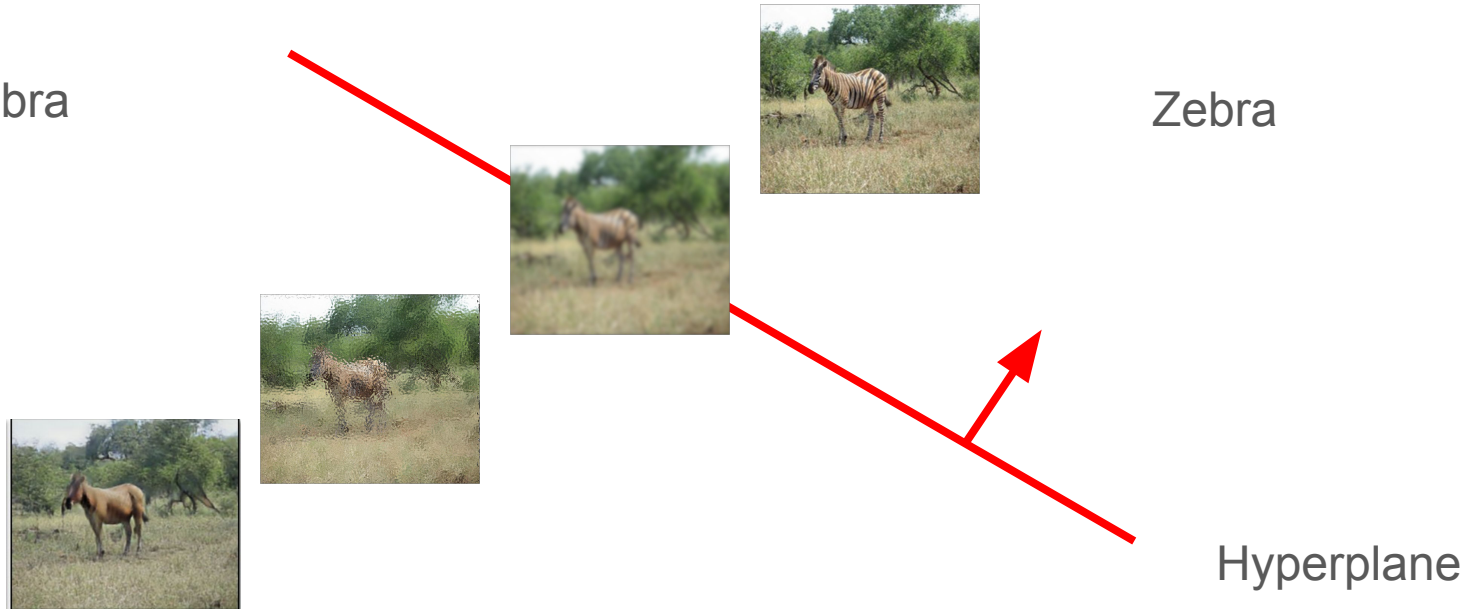
Q: The Cycle Consistency Loss is clever, but what trivial solution will you find if you minimize it alone?

Translating a zebra to a horse and then back to a zebra should recover the original image

Zhu, Jun-Yan, et al. "Unpaired image-to-image translation using cycle-consistent adversarial networks." ICCV 2017.

Classifier (Discriminator) to the rescue!

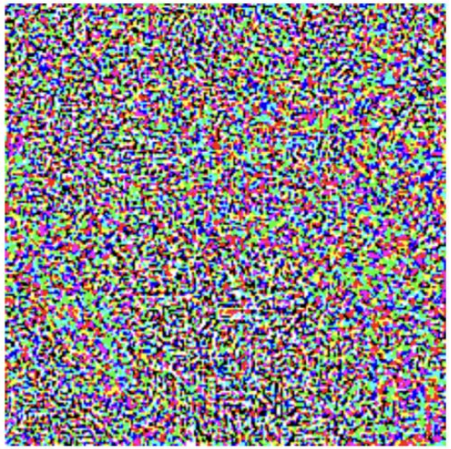
- A binary classifier that determines whether a given image is a zebra or not
- Can actually use as a learning signal!



But: Fooling classifiers is too easy

“panda”
57.7% confidence

$+ .007 \times$



Imperceptible
Modification

$=$

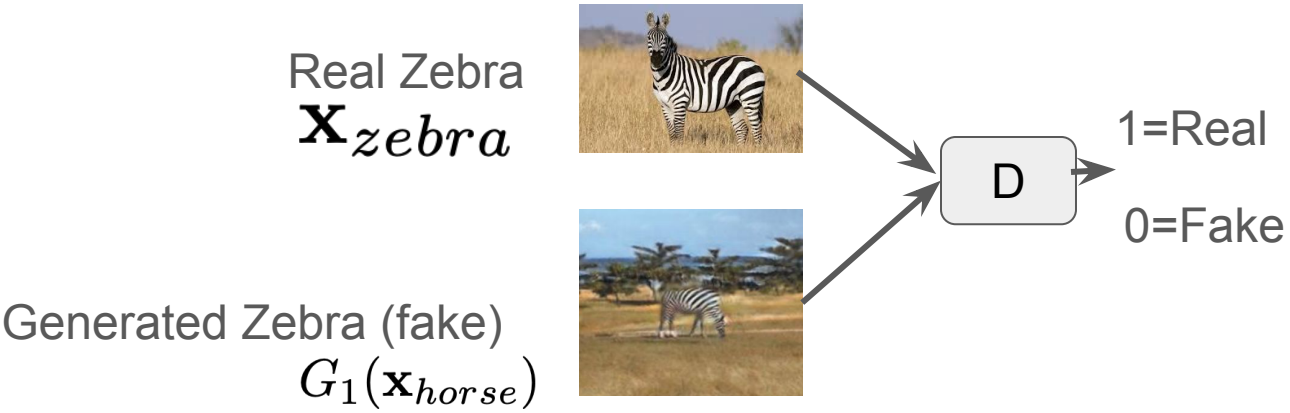
“gibbon”
99.3 % confidence

Joined training is very important!! If the discriminator D was fixed, the generator would just learn to exploit its weaknesses.

https://www.digica.com/blog/how-to-fool-a-neural-network.html

Train Discriminator and Generator jointly

$$\max_{G_1} \min_{D} \left[-\log D(\mathbf{x}_{zebra}) + \log \left(D(G_1(\mathbf{x}_{horse})) \right) \right]$$

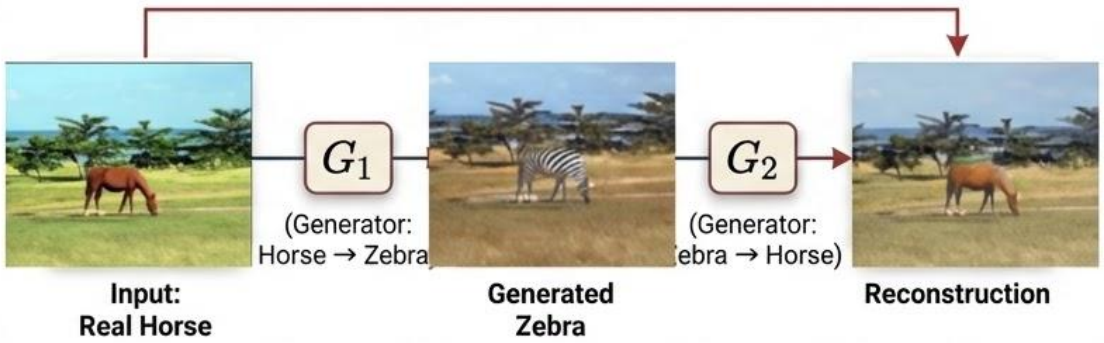


CycleGAN

Cycle Consistency & Reconstruction

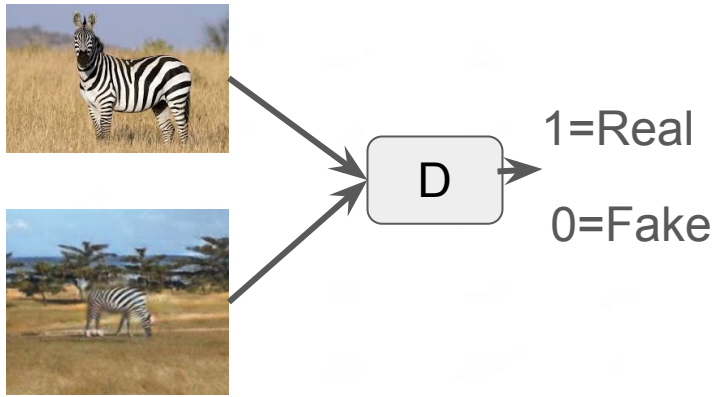
$$\|\mathbf{x}_{\text{horse}} - G_2(G_1(\mathbf{x}_{\text{horse}}))\|_2^2$$

Reconstruction Loss: Squared error between original and reconstructed image, enforcing cycle consistency.



Discriminator

$$\max_{G_1} \min_{D} \left[-\log D(\mathbf{x}_{zebra}) + \log\left(D(G_1(\mathbf{x}_{horse}))\right)\right]$$



Total Loss = Reconstruction Loss + λ Discriminator Loss

Zhu, Jun-Yan, et al. "Unpaired image-to-image translation using cycle-consistent adversarial networks." ICCV 2017.

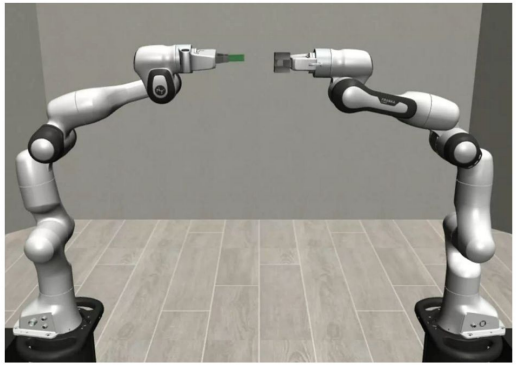
Cornell Bowers CIS

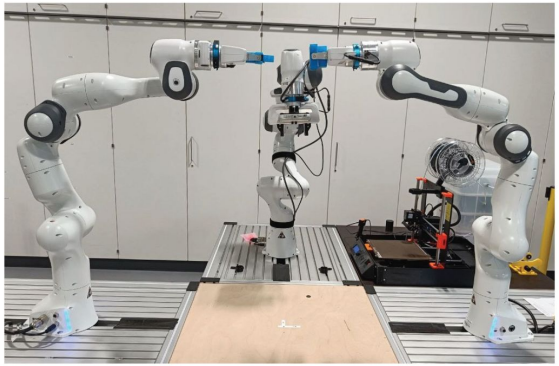
The Power of Unpaired Translation

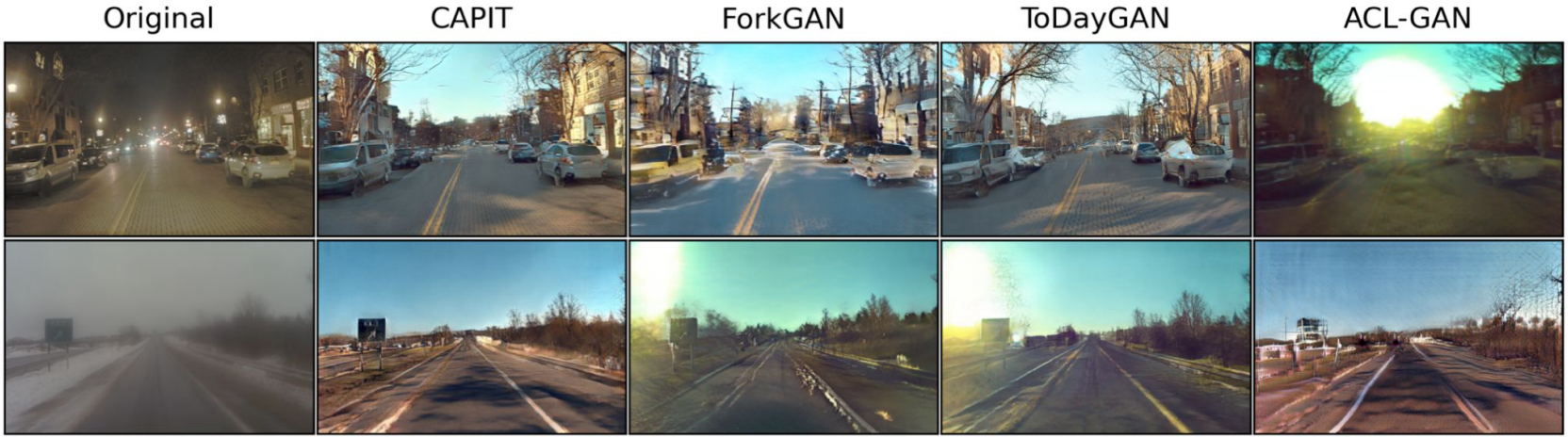
Zhu, Jun-Yan, et al. "Unpaired image-to-image translation using cycle-consistent adversarial networks." Proceedings of the IEEE international conference on computer vision. 2017.

CycleGAN Project Page

Sim2Real
Weather translation







https://www.mdpi.com/2075-1702/12/10/682

Image-to-image translation for autonomous driving from coarsely-aligned image pairs [Xia et al. 2023]

Image Transformations with Deep Learning: Four Key Cases

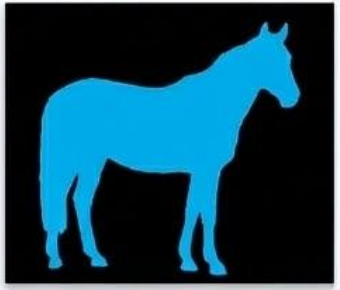
1. Image Classification

→ Output Label: “Horse”

(Input: Image → Output: Label)

2. Paired Image Translation

→



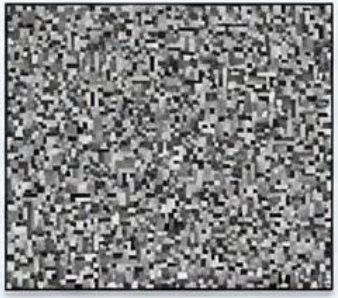
(Input: Image → Output: Image)

3. Unpaired Image Translation

→

(Input: Image → Output: Image)

4. Image Generation

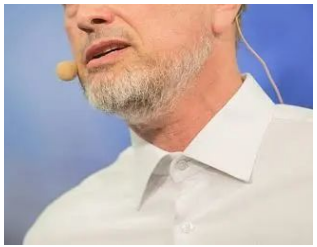


→

(Input: Noise → Output: Image)

Generative Adversarial Networks are Special Cases of Artificial Curiosity (1990) and also Closely Related to Predictability Minimization (1991)

Jürgen Schmidhuber
The Swiss AI Lab, IDSIA, USI & SUPSI, Manno-Lugano
NNAISENSE, Lugano, Switzerland
Preprint (accepted by Neural Networks, Volume 127, July 2020, Pages 58-66)



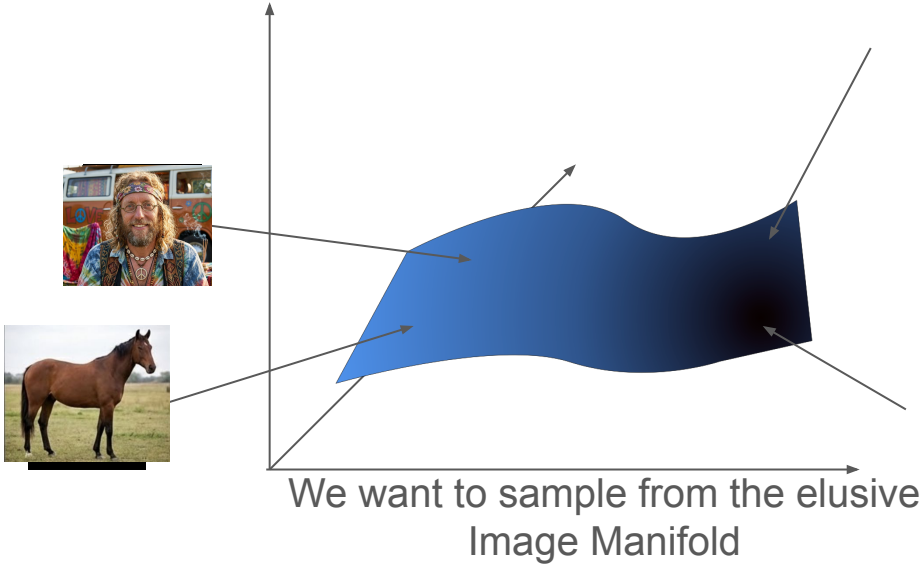
Can we directly generate something from noise?

Sample noise → sample images!

CycleGAN Project Page

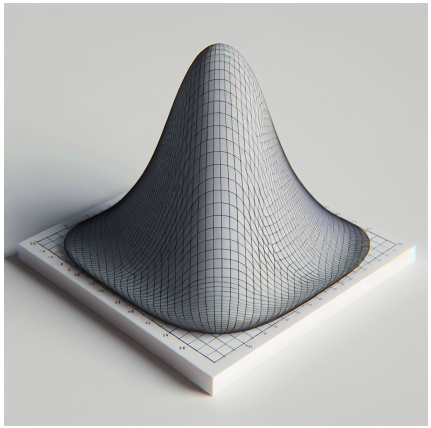
Data Manifold

- Data distribution **P(X)** defines a manifold of valid images
- Problem: data manifold takes up **tiny** volume of ambient space
- Naive random samples (e.g. within $[0,1]^d$) are always **off manifold**
- Solution: Sample from a Gaussian, then learn mapping to manifold



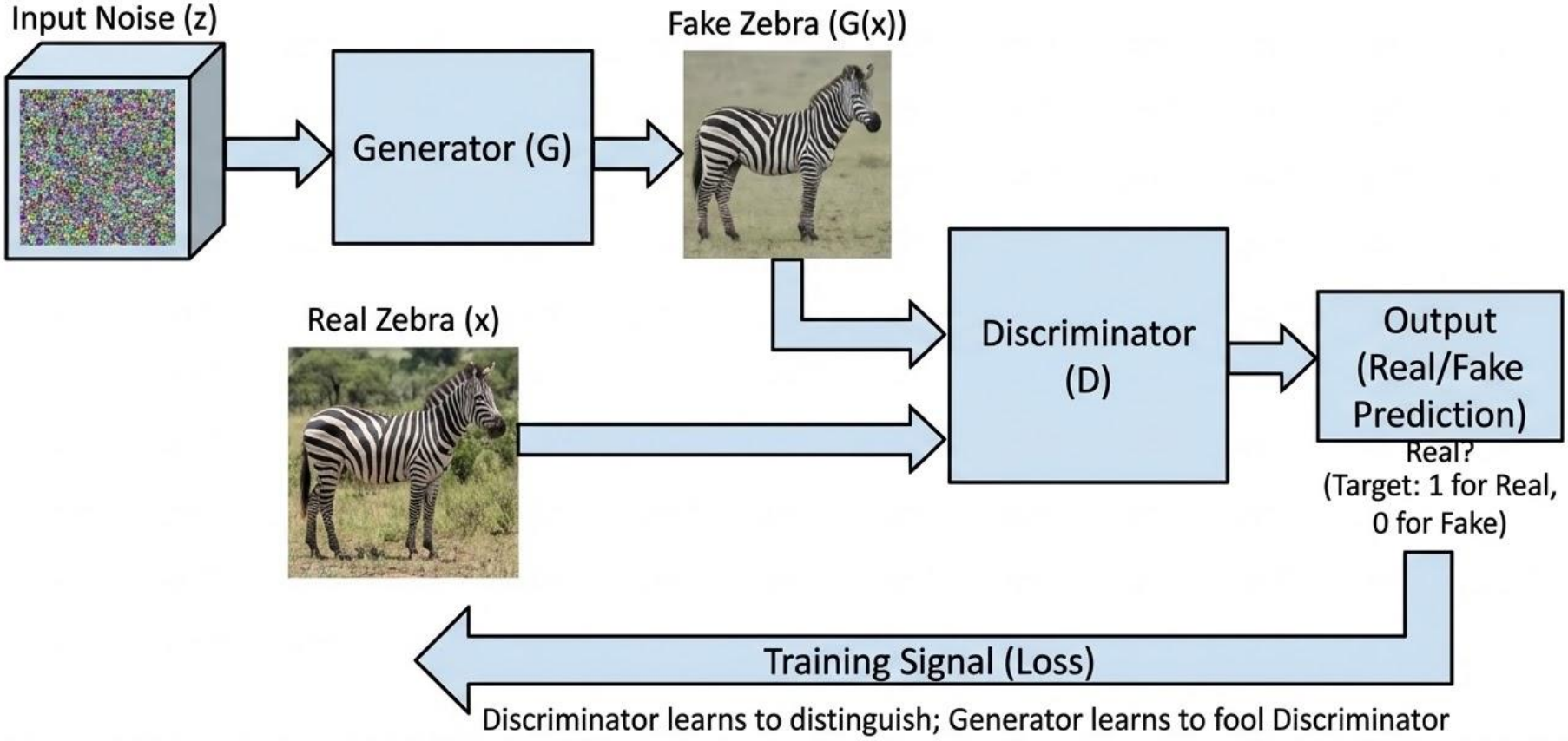
We want to sample from the elusive Image Manifold

GAN

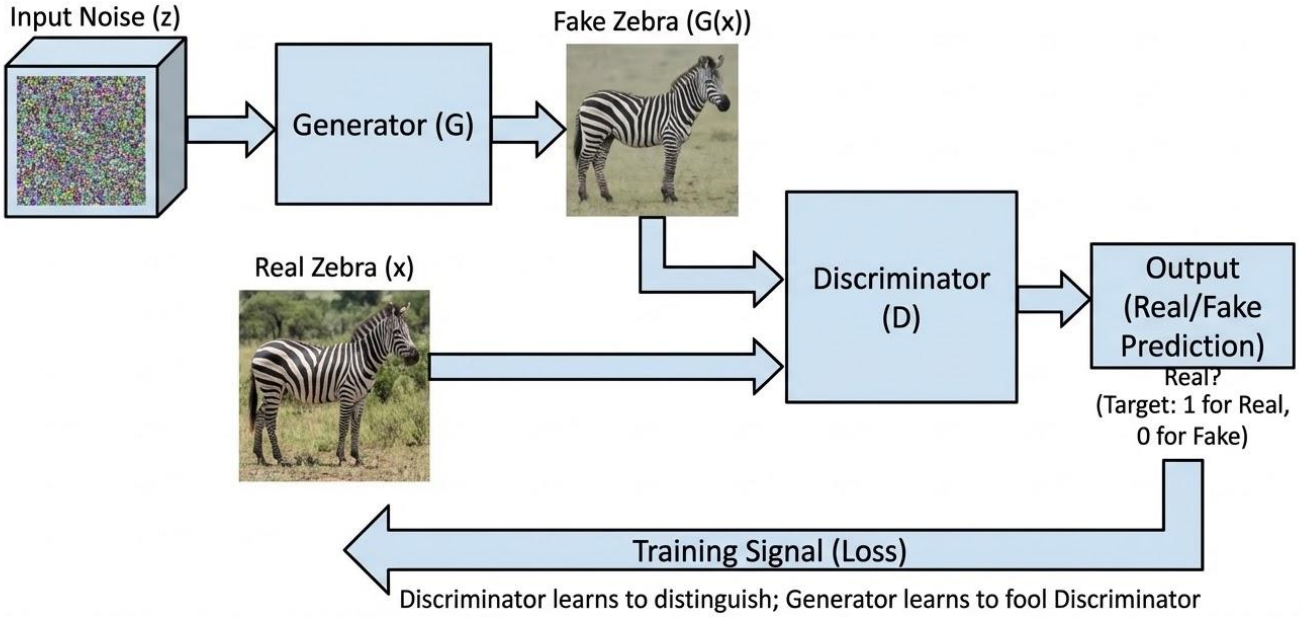


We **can** sample from a Gaussian Distribution

GAN Setup: Generating Zebras from Noise



GAN Setup: Generating Zebras from Noise



Discriminator:
Detect fake images

Generator:
Fool discriminator

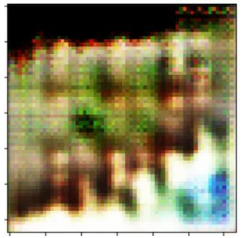
We train the generator and discriminator jointly!

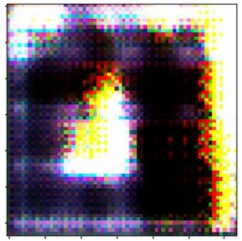
$$\max_G \min_D \left[-\log D(\mathbf{x}_{real}) + \log \left(D(G(\mathbf{z})) \right) \right]$$

Example

Real dogs

Generated





Discriminator Loss

Generally Decrease

Ideal: Confused

Training epoch

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AdaIN: Conditional GAN $P(X|s)$

Adaptive Instance Normalization (AdaIN)

A normalization layer used in generative models (notably StyleGAN) to inject style information into feature maps.

The idea is simple:

1. Normalize the feature map to remove its original statistics.
2. Apply new mean and variance derived from a style vector.

$$\text{AdaIN}(\boldsymbol{x}, \boldsymbol{s}) = \sigma(\boldsymbol{s}) \left(\frac{\boldsymbol{x} - \mu(\boldsymbol{x})}{\sigma(\boldsymbol{x})} \right) + \mu(\boldsymbol{s})$$

Where:

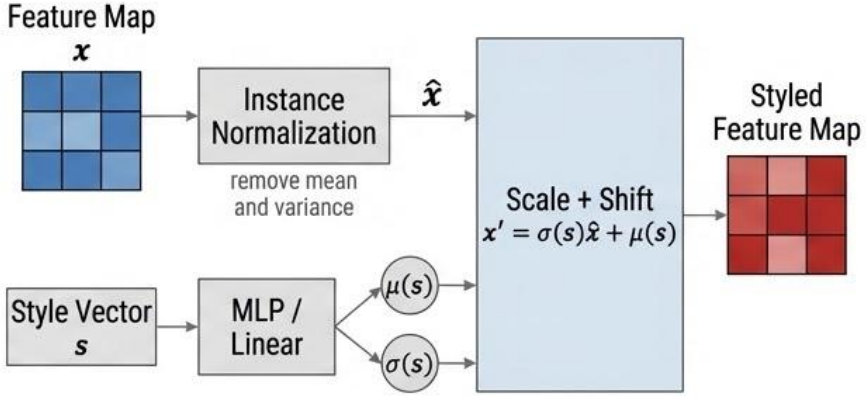
- $\boldsymbol{x}$ = feature map
- $\mu(\boldsymbol{x}), \sigma(\boldsymbol{x})$ = mean and standard deviation of the feature map (per channel)
- $\boldsymbol{s}$ = style vector
- $\mu(\boldsymbol{s}), \sigma(\boldsymbol{s})$ = learned scale and shift parameters produced from $\boldsymbol{s}$

Intuition

- First remove the existing feature statistics.
- Then impose style-dependent statistics.

Thus the style vector controls how the features look (texture, colors, shapes), while the underlying features still encode structure.

Example Illustration



Visual intuition:

Feature map before AdaIN:

After AdaIN with style:

- Structure stays
- appearance statistics change

AdaIN normalizes feature maps and then re-injects **style-dependent** statistics, allowing the **generator to control appearance** through a **style vector**.

GANs

- First generative model capable of realistic high-resolution image synthesis
- Very fast generation!

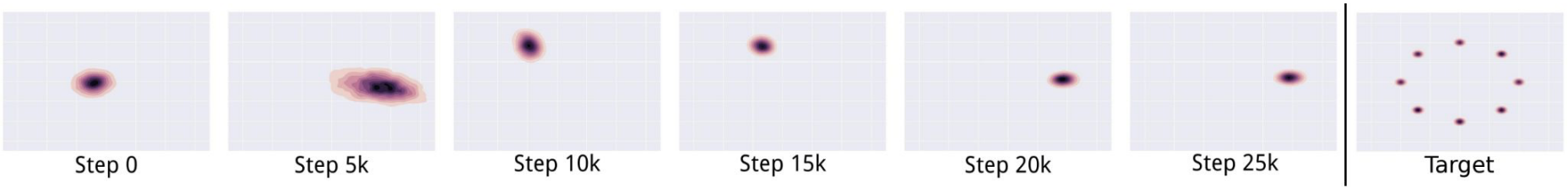
Latent Space Properties

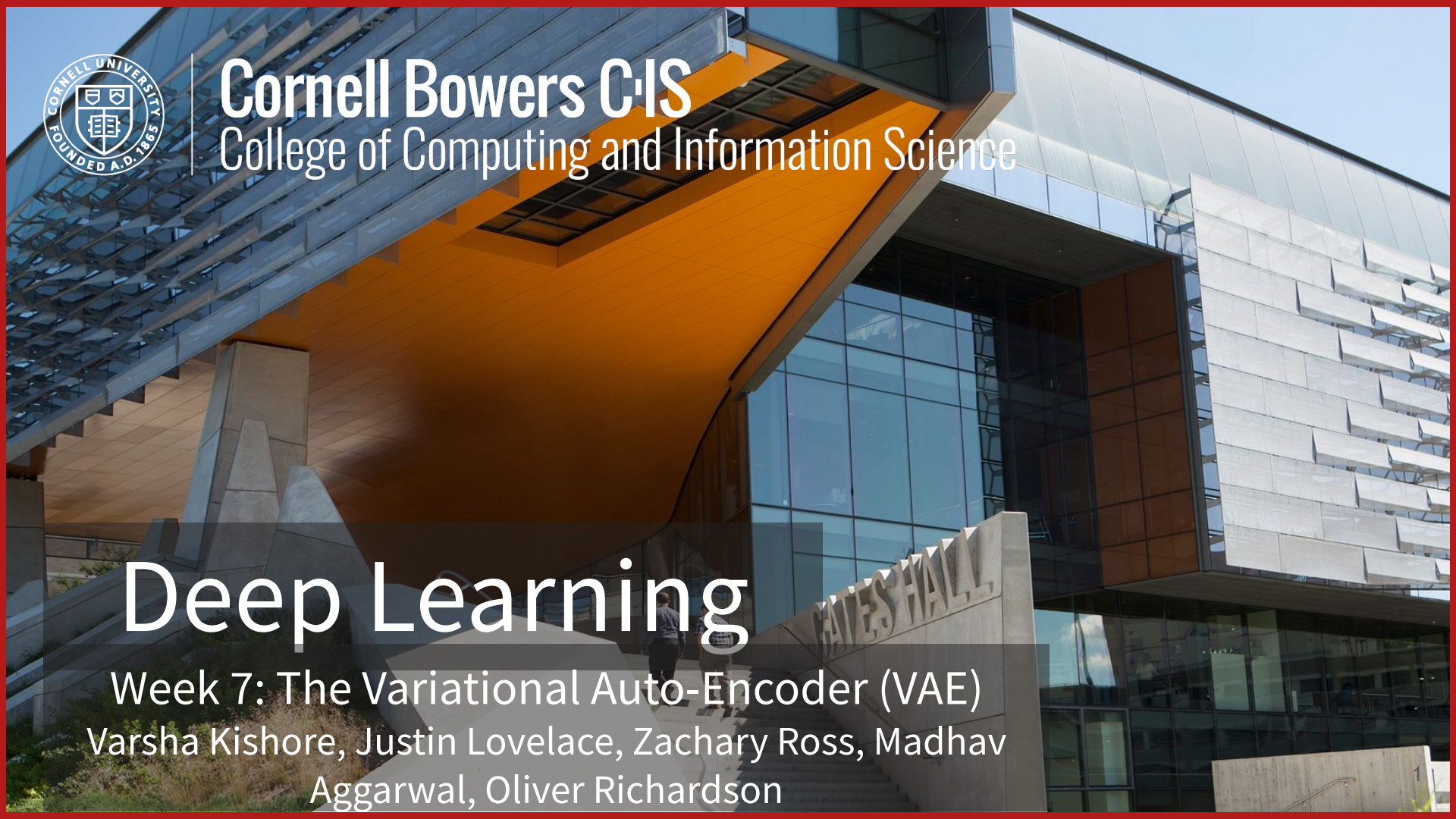
Gaussian noise vector is meaningful

- Similar noise vectors lead similar images
- Noise dimensions are meaningful!
- Can exploit this to control generation

Limitations of GANs

- Consider training a GAN on a dataset of dogs and cats
  - Generator could specialize in generating realistic dogs
  - Successfully fools the discriminator!
- Minimax training objective is hard to optimize!
  - Can lead to oscillations/instability during training
- Not guaranteed to converge to a good solution
  - Sensitive to hyperparameter settings, network architectures, and the choice of loss functions
- Mode Collapse is real!





Deep Learning

Week 7: The Variational Auto-Encoder (VAE)

Varsha Kishore, Justin Lovelace, Zachary Ross, Madhav Aggarwal, Oliver Richardson

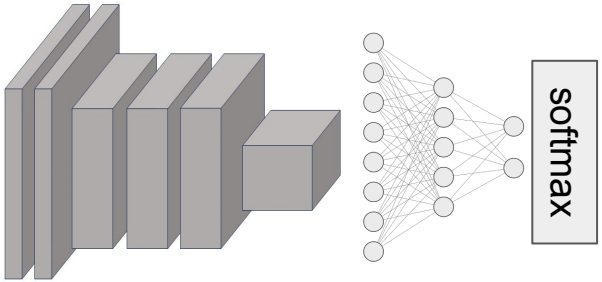
Discriminative Models

typically supervised

Goal: model $p(Y|X)$

from samples of p(X,Y)

(* so that we can predict most likely labels)



Generative Models

unsupervised

Goal: model $p(X)$

from samples of p(X)

(* so that we can generate artificial/new data)

Examples:

- GANs + variants
- Normalizing Flow Models
- Variational Autoencoders
  - Diffusion Models

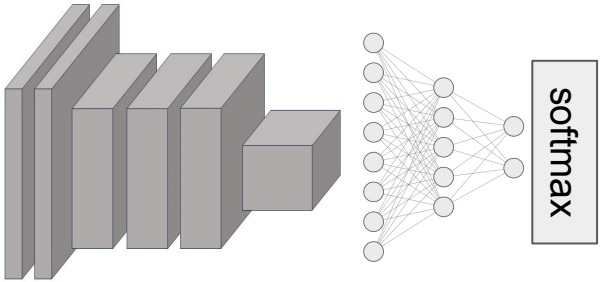
Discriminative Models

typically supervised

Goal: model $p(Y|X)$

from samples of p(X,Y)

(* so that we can predict most likely labels)



Generative Models

(Conditional generation)

Goal: model $p(X|Y)$

from samples of p(X,Y)

(* so that we can generate artificial/new data)

Examples:

- GANs + variants
- Normalizing Flow Models
- Variational Autoencoders
  - Diffusion Models

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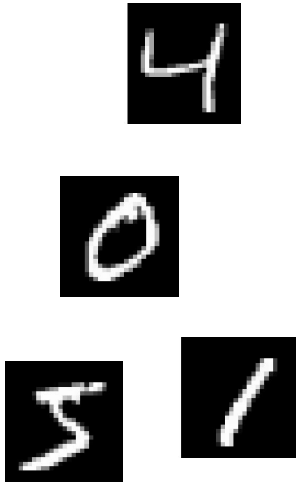
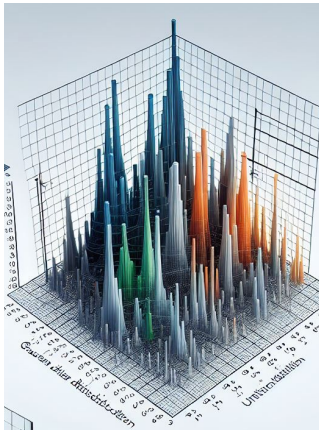
Big Picture

Data sampled from true (but elusive) P(X)



Learn approximate data distribution

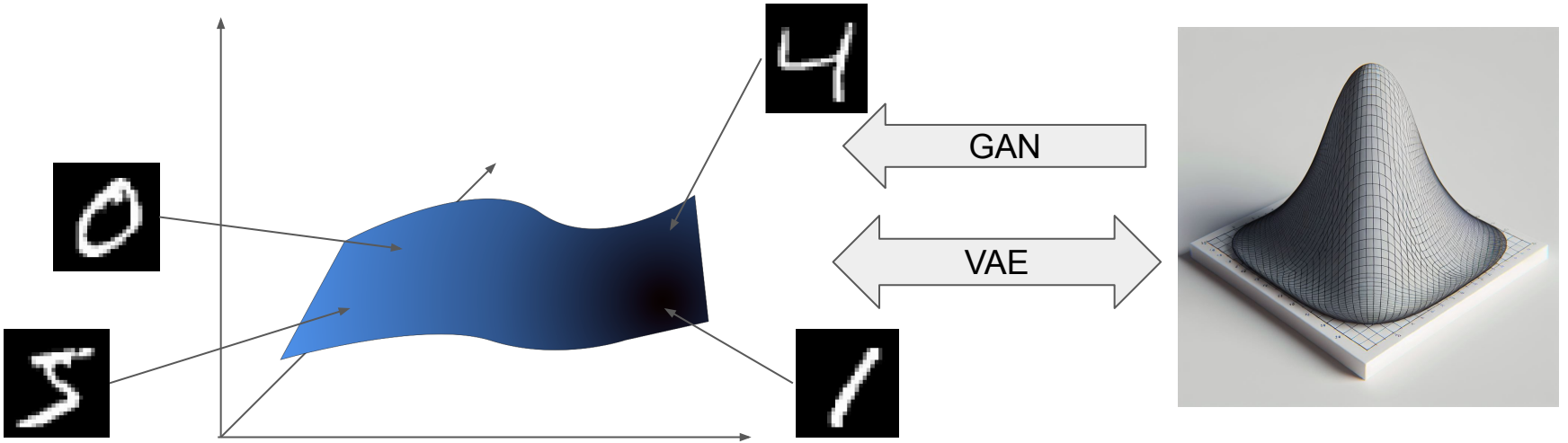
Q(X)≈P(X)



New (fake) data drawn from Q(x)

Data Manifold

- Data distribution **P(X)** defines a manifold of valid images
- Problem: data manifold takes up **tiny** volume of ambient space
- Naive random samples (e.g. within $[0,1]^d$) are always **off manifold**
- Solution: Sample from a Gaussian, then learn mapping to and from manifold

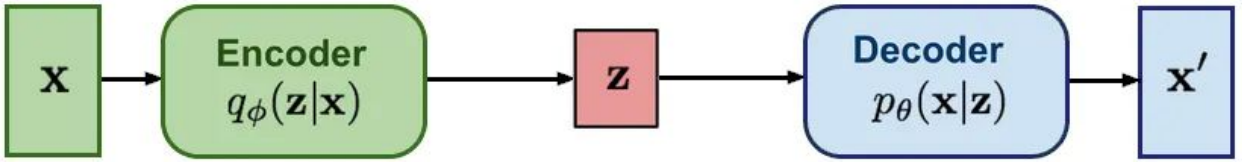


We want to sample from the elusive Image Manifold

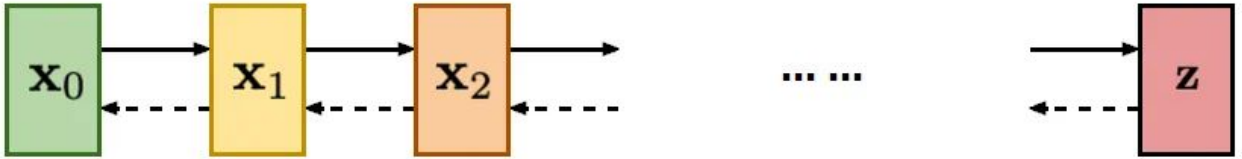
We **can** sample from a Gaussian Distribution

GAN: Adversarial training

VAE: maximize variational lower bound



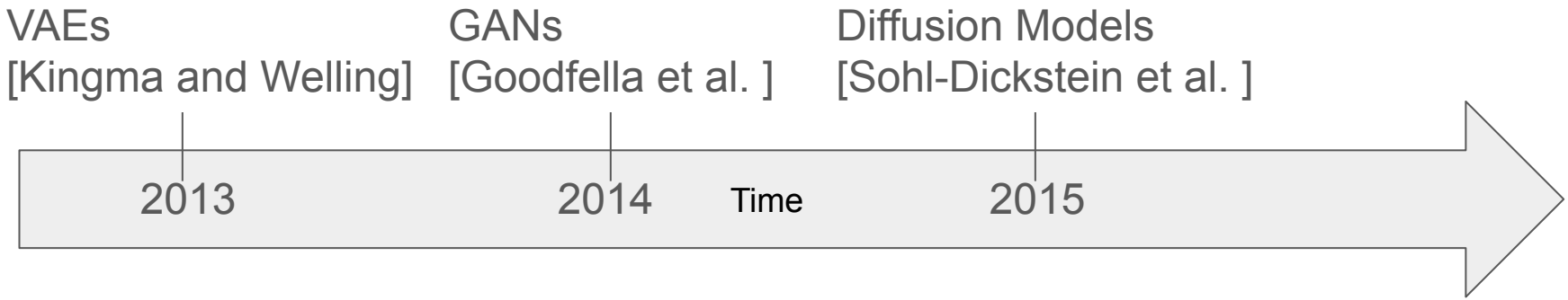
Diffusion models: Gradually add Gaussian noise and then reverse



https://miro.medium.com/v2/resize:fit:1400/format:webp/1*_5GpdejeOvt61ew4aPtT_g.png

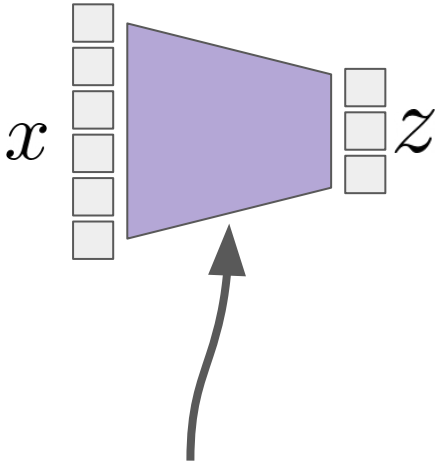
Timeline

- VAEs preceded GANs.
- In fact, GANs were motivated to fix some of the problems of VAEs.
- VAEs are important to understand **Diffusion Models**



Dimensionality Reduction

Data is typically from a **low dimensional** distribution embedded in a much **higher dimensional** ambient space.



Want to map images $x \in \mathbb{R}^D$
to low-dimensional $z \in \mathbb{R}^d$

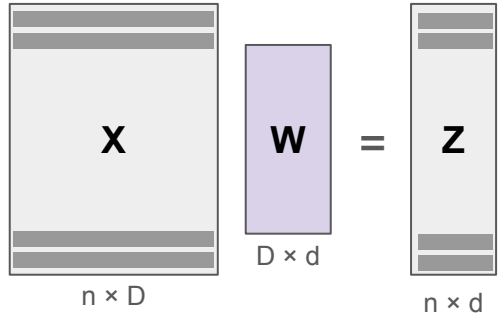
Often for the purposes of

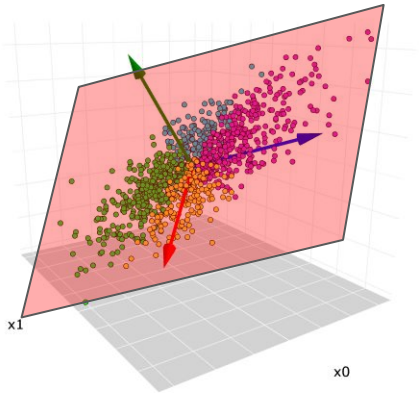
- visualization
- extracting important features (for downstream tasks)
- representing meaningful relationships between samples
- In this lecture sampling!

Question: what properties should this mapping have?

Principal Component Analysis (PCA)

- **assumption: data manifold is a subspace**
- $\mathbf{z} = \mathbf{W}(\mathbf{x} - \mu)$ (a linear transformation)
- $\mathbf{x} \approx \mathbf{W}^\top \mathbf{z} + \mu$ (reconstruction)
- capture as much variance as possible



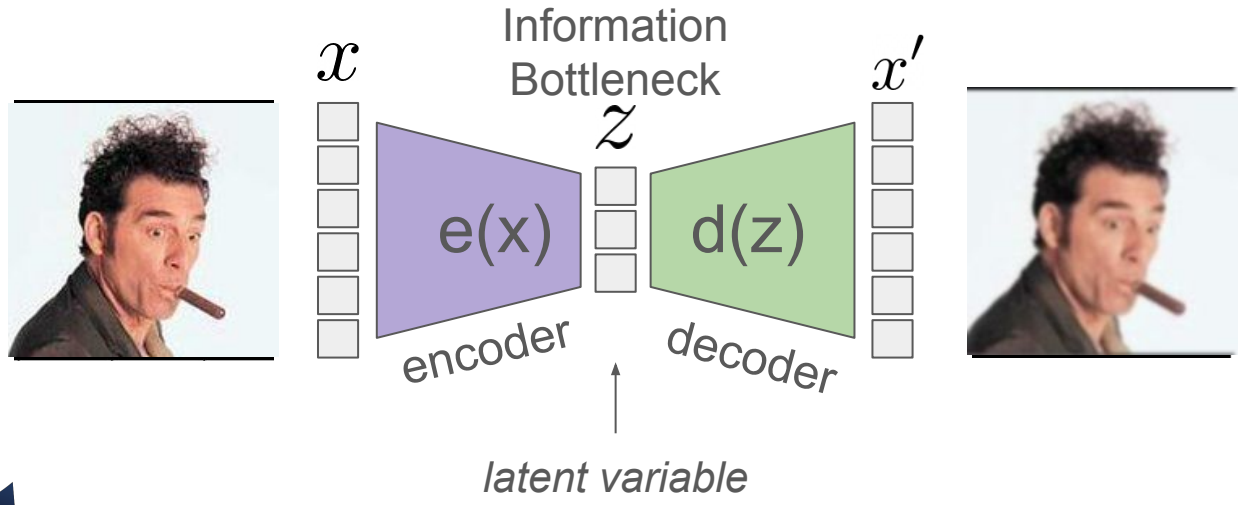


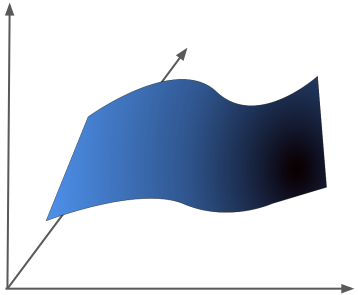
Can be computed directly with linear algebra: take leading eigenvectors of (centered) scatter matrix XX'!

https://towardsdatascience.com/principal-component-analysis-pca-explained-visually-with-zero-math-1cbf392b9e7d/

Autoencoders [Kramer, 1991]

Non-linear dimensionality reduction.



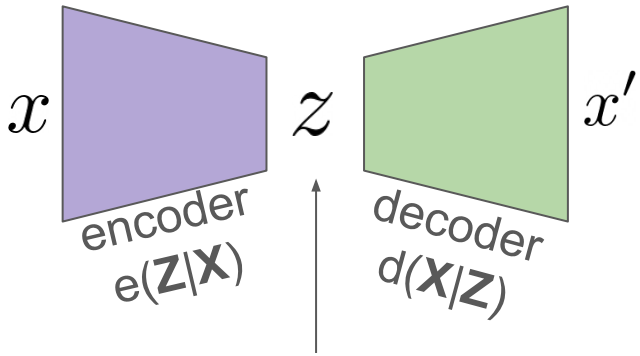


Question: What loss function should we use to learn e() and d()?
What happens if e(x) and d(z) are both linear functions?

Autoencoders [Kramer, 1991]

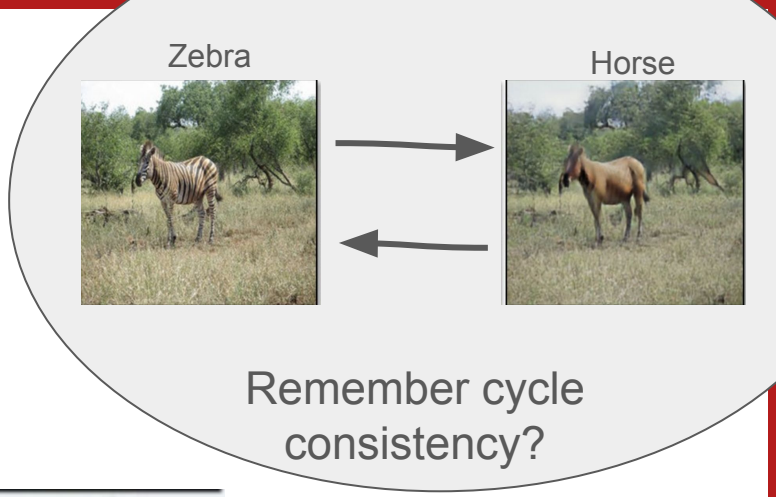
Typical loss: Squared loss, or absolute loss

Information Bottleneck

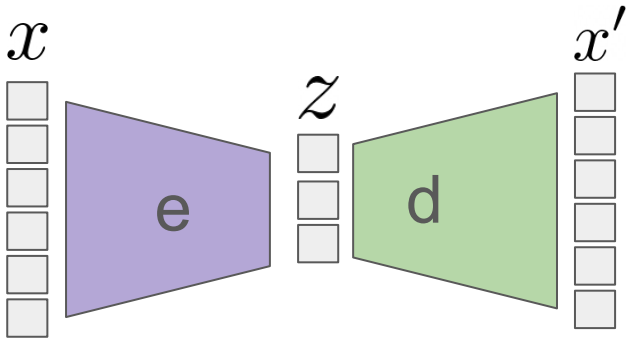


$$\min_{\phi,\theta} \sum_{\mathbf{x}\in\mathcal{D}} (\mathbf{x} - \mathbf{x}')^2$$

$$\mathbf{x}' = d_\theta(e_\phi(\mathbf{x}))$$

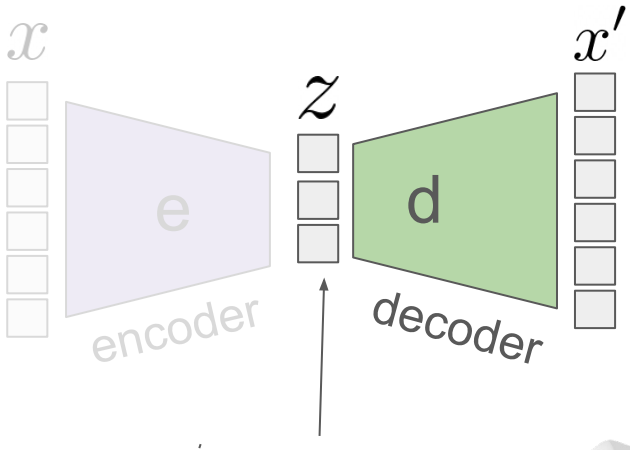


Idea: Sampling from a trained Autoencoder



- GANs train the decoder with a discriminator
- VAEs ensure quality with
  - Reconstruction loss
  - KL regularization (in a few slides)

Idea: Sampling from a trained Autoencoder

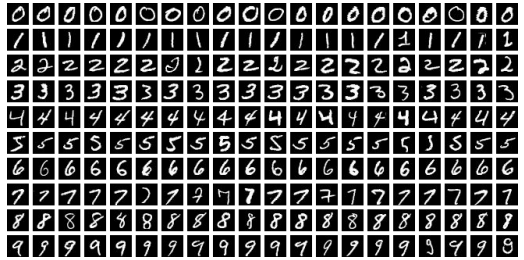


- GANs train the decoder with a discriminator
- VAEs ensure quality with
  - Reconstruction loss
  - KL regularization (in a few slides)

Feed in noise, sampled from some distribution P(z)

<u>Crucial insight:</u>
We can amend latent space so that it is easy to sample from it.

Autoencoder trained on MNIST: latent space



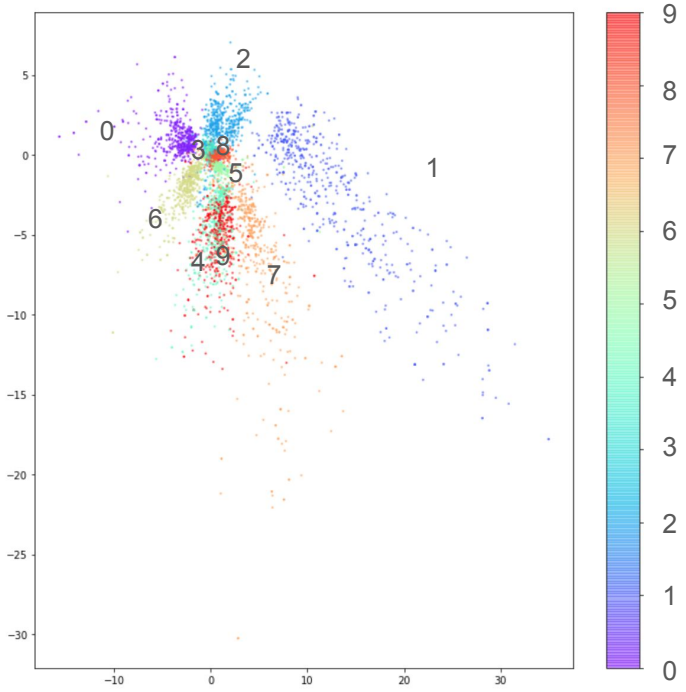


Figure 3-8. Plot of the latent space, colored by digit

Naive representation (without any special effort), not favorable:

- lots of empty space
- no symmetries between digit representations
- **Not easy to sample in latent space**

Naive sampling in latent space does not work

reconstructed sample

$$x' = d(e(x))$$

new image?

$$x' = d(\text{ noise })$$

Some Fundamentals

of probability and information

Building Blocks:

- Conditional and marginal probabilities
- Surprisal / Negative Log Likelihood
- Relative Entropy / KL Divergence

Conditional and Marginal Probabilities

$$p(X, Y) = p(Y|X)p(X)$$

$$p(X) = \int p(X, y)\,\mathrm{d}y$$

Equation (2): Quick 60s Stats Puzzle

Prove that for any random variable X,Z.

$$p(\mathbf{x}) = \frac{p(\mathbf{x}, \mathbf{z})}{p(\mathbf{z}|\mathbf{x})}$$

Hint: Later this rule will come in handy. Remember it as “Equation (2)”.

KL Divergence (a.k.a. relative entropy)

$$\boldsymbol{D}(p \,\|\, q) := \mathop{\mathbb{E}}_{x \sim p} \left[\log \frac{p(x)}{q(x)} \right]$$

reality (e.g., dataset)

model

$$\mathbb{E}_{x \sim p} [\log p(x) - \log q(x)]$$

likelihood in real world; does not depend on model q

Likelihood under model

- non-negative $\boldsymbol{D}(p \,\|\, q) \geq 0$
- zero means same $\boldsymbol{D}(p \,\|\, q) = 0 \iff p = q$
- not symmetric
- has many other, uniquely nice properties …

KL Divergence

Wei-Chiu's Coin

Kilian's Coin

× 2

Question:
What is higher **KL(Wei-Chiu || Kilian)** or **KL(Kilian || Wei-Chiu)**?

[link to visualization]

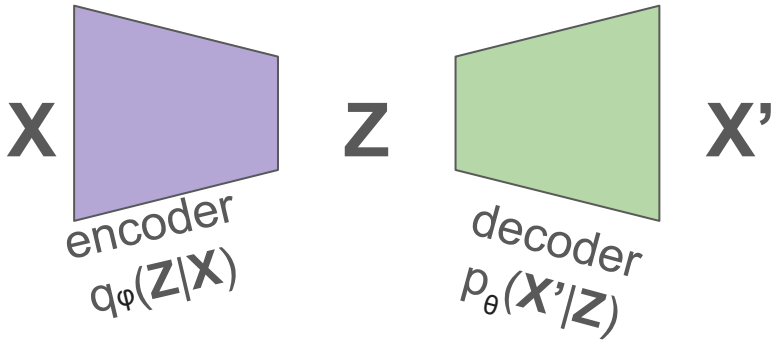
Building Blocks:

✅ Conditional and marginal probabilities
✅ Surprisal / Negative Log Likelihood
✅ Relative Entropy / KL Divergence

VAEs,
Step 1: Make AutoEncoder probabilistic

Reconstruction Loss, using surprisal

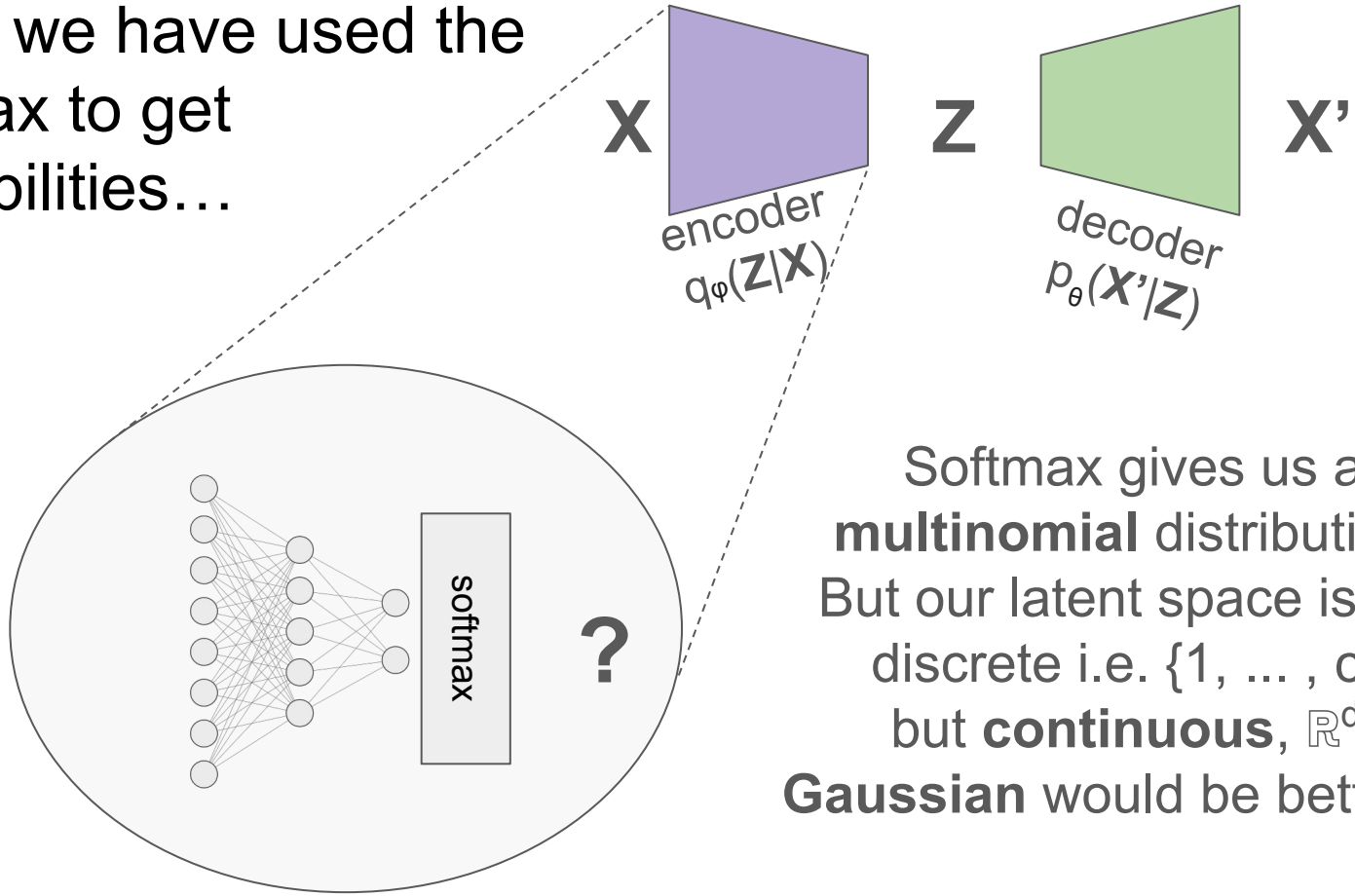
Back to our AutoEncoder, but this time we make everything **probabilistic**!



$$\max_{\phi,\theta} \mathbb{E}_{z\sim q_\phi(\mathbf{z}|\mathbf{x})}[\log(p_\theta(\mathbf{x}|\mathbf{z}))]$$

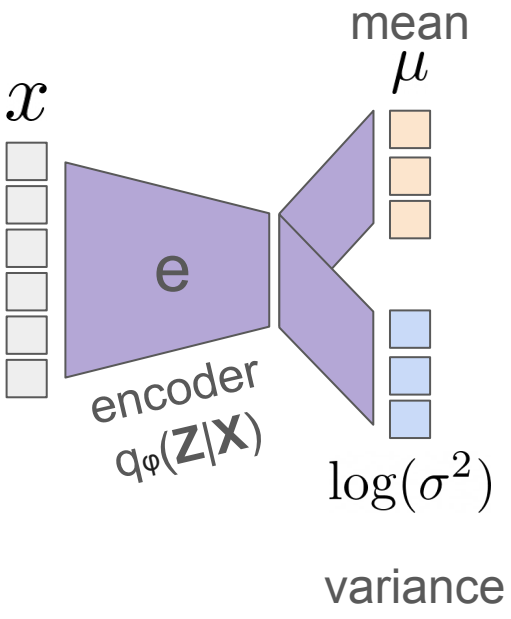
How likely would it be to encode x, decode the result, and recover x?

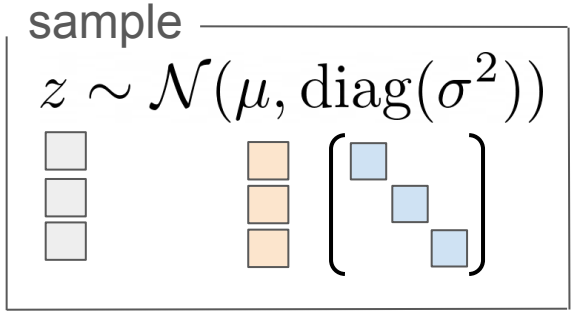
So far we have used the softmax to get probabilities…

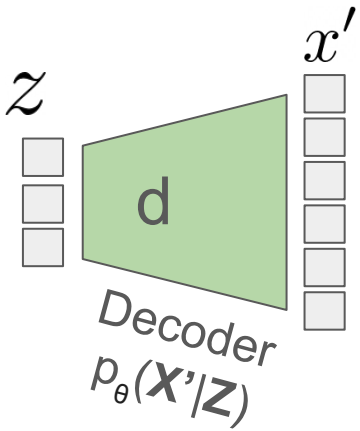


Softmax gives us a **multinomial** distribution. But our latent space is not discrete i.e. {1, ... , c}, but **continuous**, $\mathbb{R}^d$! **Gaussian** would be better …

Probabilistic **Encoder** (Gaussian)







Problem: backpropagation through sampling process?

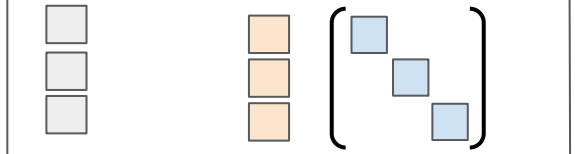
$$\max_{\phi,\theta} \mathbb{E}_{z\sim q_\phi(\mathbf{z}|\mathbf{x})}[\log\left(p_\theta(\mathbf{x}|\mathbf{z})\right)]$$

The Reparameterization Trick

$$\mathcal{N}(\mu, \text{diag}(\sigma^2)) = \mu + \sigma \odot \mathcal{N}(0, I)$$

sample

$z \sim \mathcal{N}(\mu, \text{diag}(\sigma^2))$



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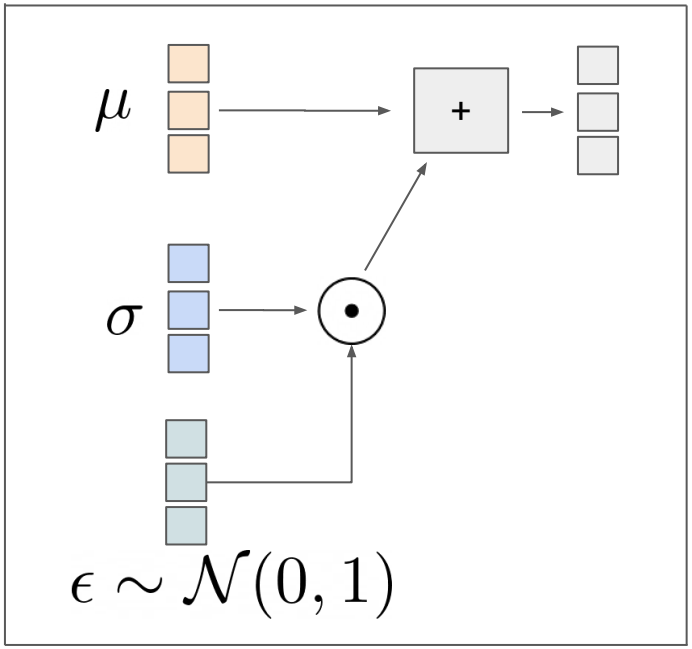
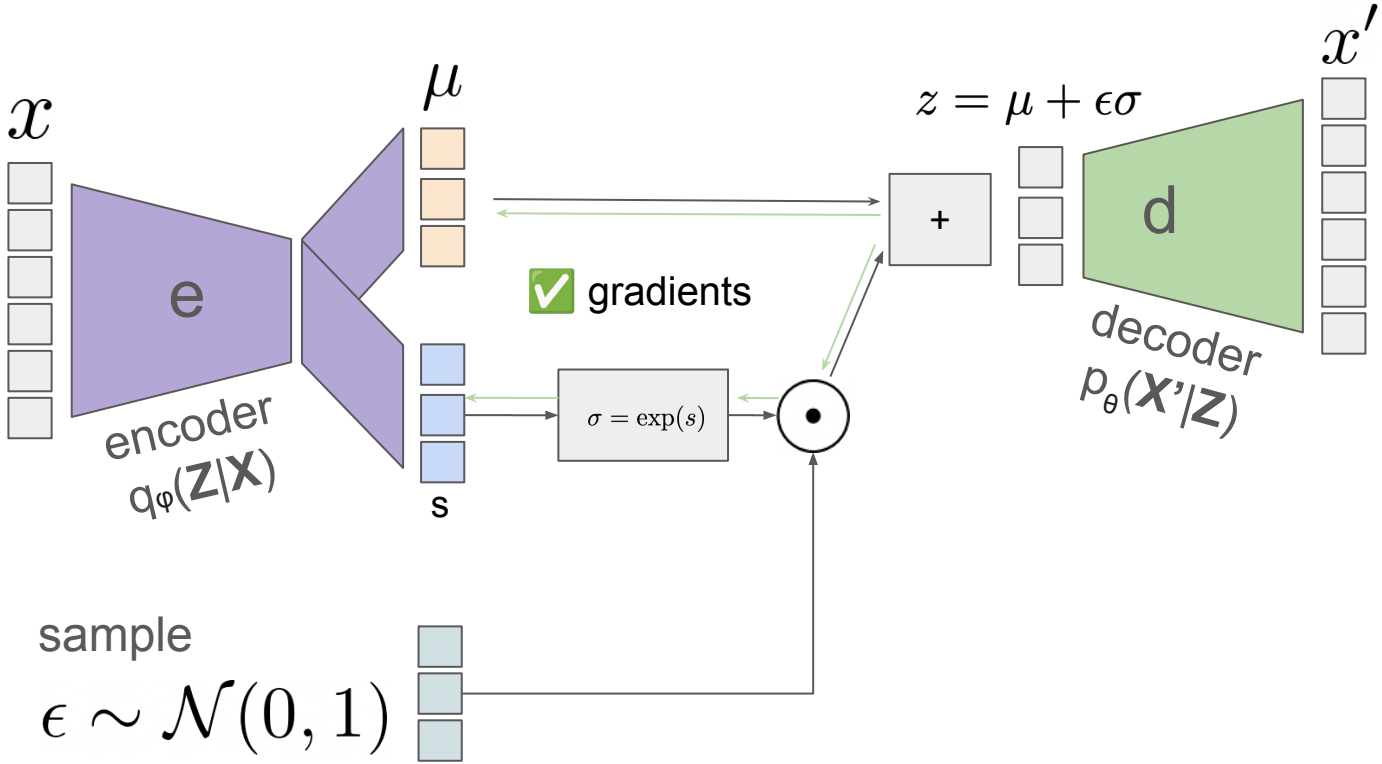
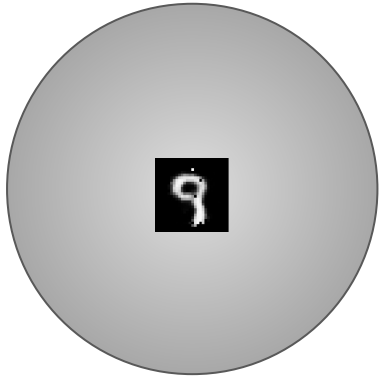
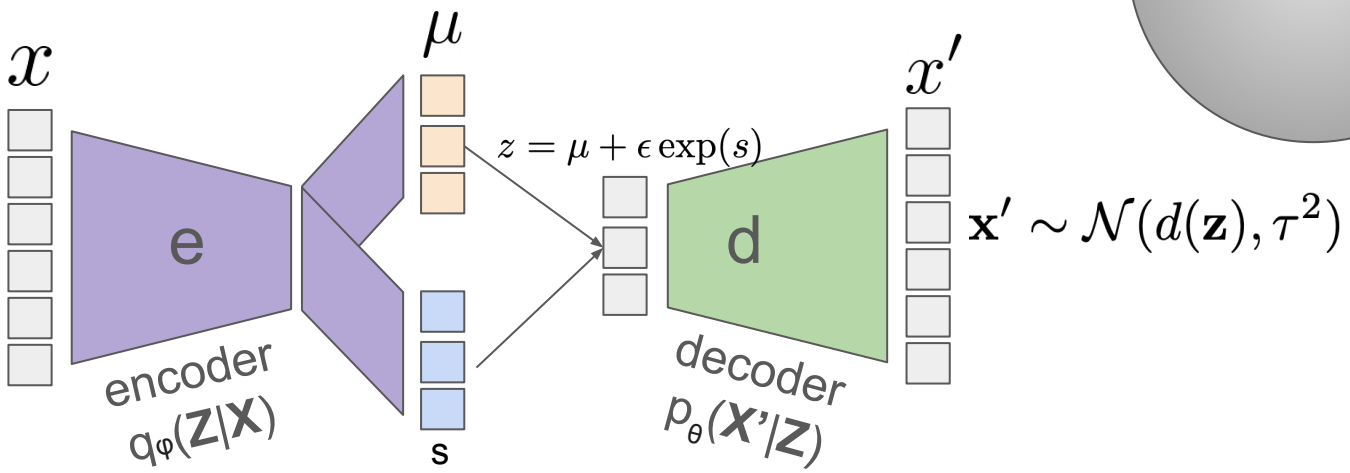


The Reparameterization Trick

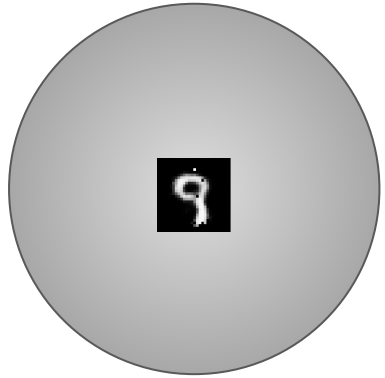


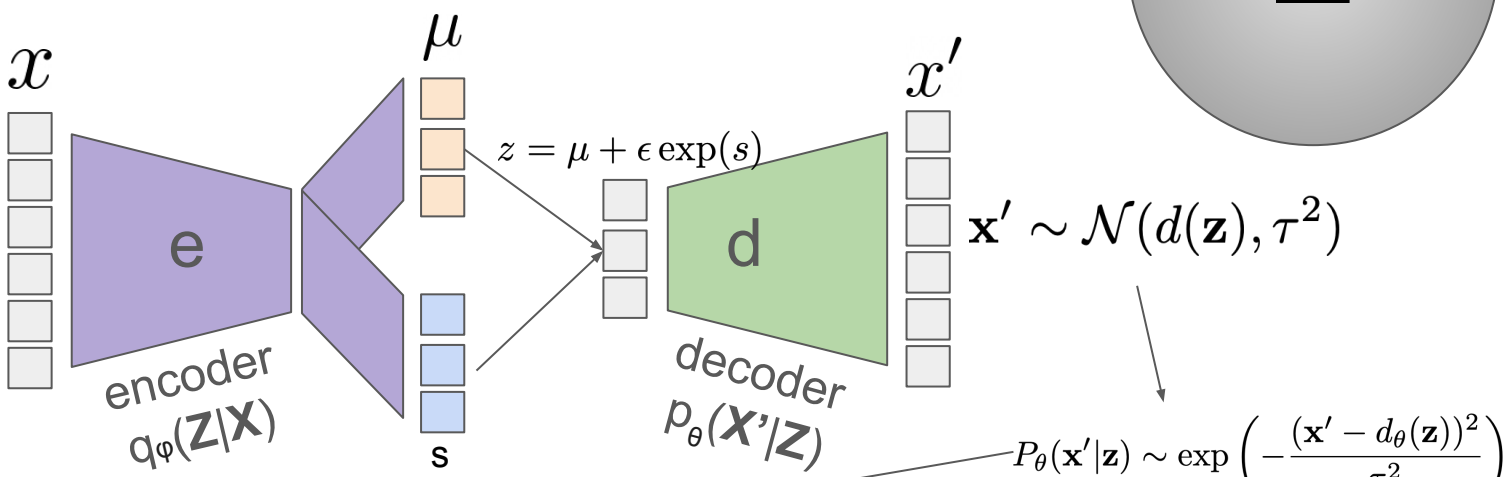
Probabilistic **decoder** (Gaussian)





Probabilistic **decoder** (Gaussian)



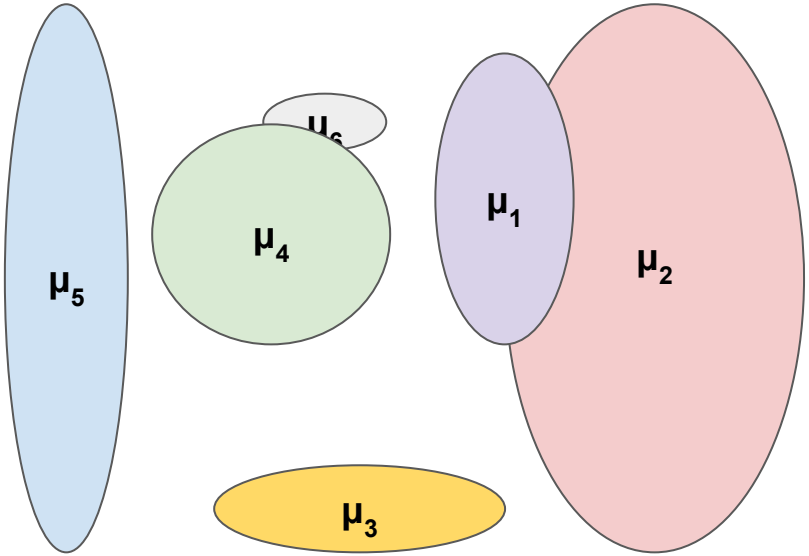


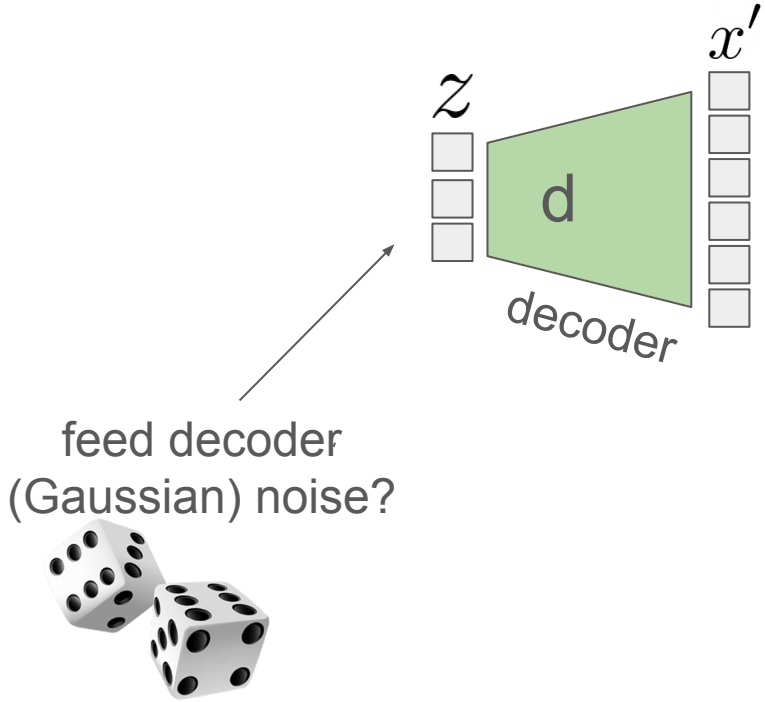
$$\max_{\phi,\theta} \mathbb{E}_{z\sim q_\phi(\mathbf{z}|\mathbf{x})}[\log(p_\theta(\mathbf{x}|\mathbf{z}))] = \min_{\phi,\theta} \mathbb{E}_{z\sim q_\phi(\mathbf{z}|\mathbf{x})}\left[(\mathbf{x} - d_\theta(\mathbf{z}))^2\right]$$

Plugging the output distribution into the reconstruction loss, results in the squared loss.

Step 2: How do we sample in latent space?

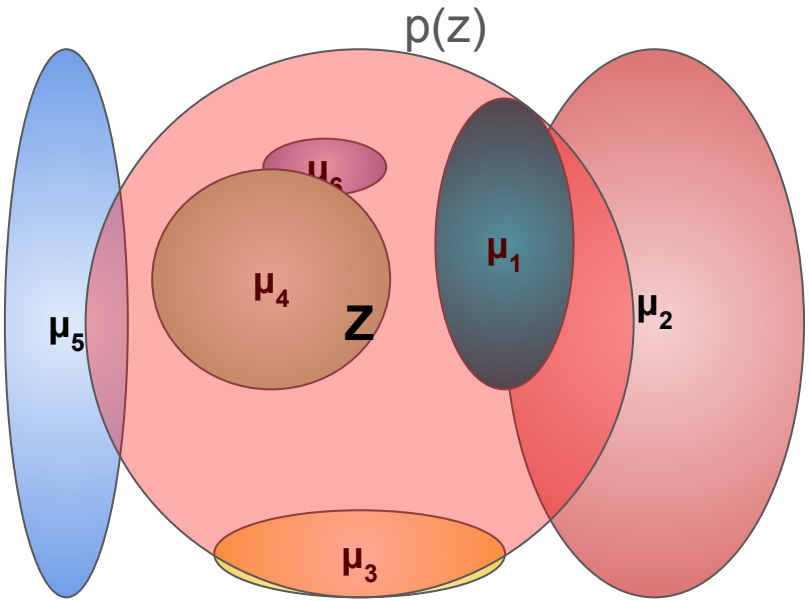
How can we sample, if each sample has its own latent distribution?





Step 2: How do we sample in latent space?

Solution: Regularize all distributions to be close to the standard normal ***N(0;I)***.



maximize

$$\underbrace{\mathbb{E}_{q_\phi(\boldsymbol{z}|\boldsymbol{x})}\left[\log p_\theta(\boldsymbol{x}|\boldsymbol{z})\right]}_{\text{reconstruction term}} - \underbrace{D_{\mathrm{KL}}(q_\phi(\boldsymbol{z}|\boldsymbol{x}) \parallel p(\boldsymbol{z}))}_{\text{prior matching term}}$$

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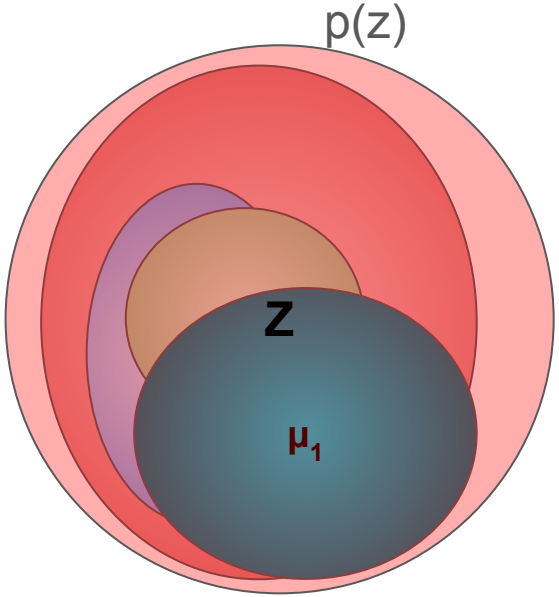

maximize

$$\underbrace{\mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\left[\log p_{\boldsymbol{\theta}}(\boldsymbol{x}|\boldsymbol{z})\right]}_{\text{reconstruction term}} - \underbrace{D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x}) \parallel p(\boldsymbol{z}))}_{\text{prior matching term}}$$

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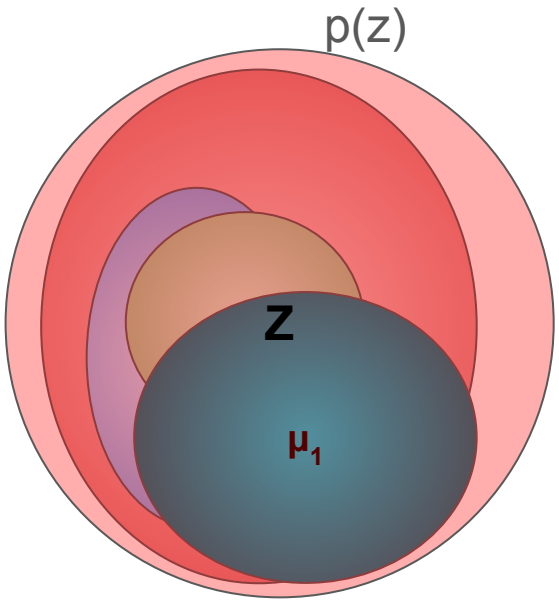
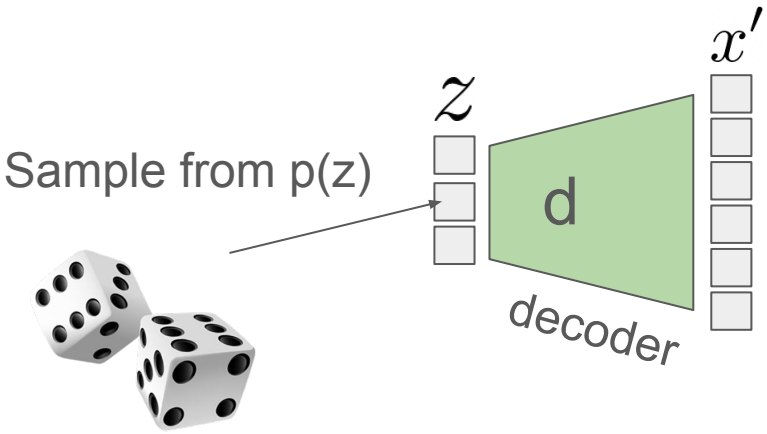


maximize

$$\underbrace{\mathbb{E}_{q_\phi(\boldsymbol{z}|\boldsymbol{x})}\left[\log p_\theta(\boldsymbol{x}|\boldsymbol{z})\right]}_{\text{reconstruction term}} - \underbrace{D_{\mathrm{KL}}(q_\phi(\boldsymbol{z}|\boldsymbol{x}) \parallel p(\boldsymbol{z}))}_{\text{prior matching term}}$$



Evidence Lower Bound (ELBO)

$$\log p(\boldsymbol{x}) = \log p(\boldsymbol{x}) \int q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x}) d\boldsymbol{z} \qquad \left(\text{Multiply by } 1 = \int q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x}) d\boldsymbol{z}\right)$$

$$= \int q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})(\log p(\boldsymbol{x})) d\boldsymbol{z} \qquad \text{(Bring evidence into integral)}$$

$$= \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\left[\log p(\boldsymbol{x})\right] \qquad \text{(Definition of Expectation)}$$

$$= \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\left[\log \frac{p(\boldsymbol{x},\boldsymbol{z})}{p(\boldsymbol{z}|\boldsymbol{x})}\right] \qquad \text{(Apply Equation 2)}$$

$$= \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\left[\log \frac{p(\boldsymbol{x},\boldsymbol{z}) q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}{p(\boldsymbol{z}|\boldsymbol{x}) q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\right] \qquad \left(\text{Multiply by } 1 = \frac{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\right)$$

$$= \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\left[\log \frac{p(\boldsymbol{x},\boldsymbol{z})}{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\right] + \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\left[\log \frac{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}{p(\boldsymbol{z}|\boldsymbol{x})}\right] \qquad \text{(Split the Expectation)}$$

$$= \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\left[\log \frac{p(\boldsymbol{x},\boldsymbol{z})}{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\right] + D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x}) \parallel p(\boldsymbol{z}|\boldsymbol{x})) \qquad \text{(Definition of KL Divergence)}$$

$$\geq \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\left[\log \frac{p(\boldsymbol{x},\boldsymbol{z})}{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\right] \qquad \text{(KL Divergence always } \geq 0)$$

[Calvin Luo https://arxiv.org/abs/2208.11970]

Evidence Lower Bound (ELBO)

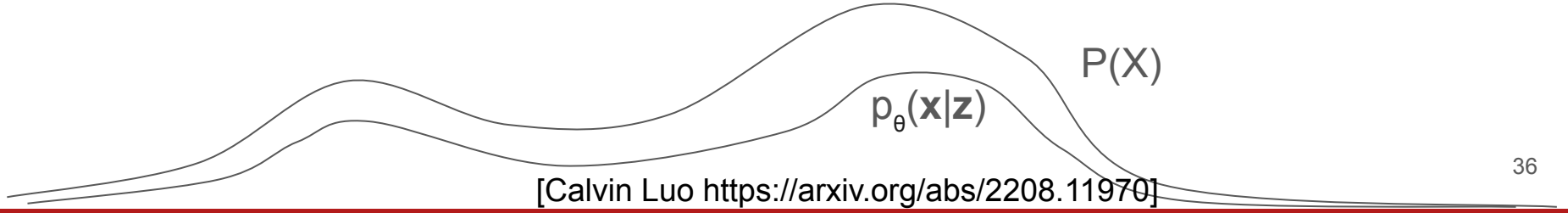
$$\mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\left[\log \frac{p(\boldsymbol{x},\boldsymbol{z})}{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\right] = \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\left[\log \frac{p_{\boldsymbol{\theta}}(\boldsymbol{x}|\boldsymbol{z})p(\boldsymbol{z})}{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\right] \quad \text{(Chain Rule of Probability)}$$

$$= \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\left[\log p_{\boldsymbol{\theta}}(\boldsymbol{x}|\boldsymbol{z})\right] + \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\left[\log \frac{p(\boldsymbol{z})}{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\right] \quad \text{(Split the Expectation)}$$

$$= \underbrace{\mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\left[\log p_{\boldsymbol{\theta}}(\boldsymbol{x}|\boldsymbol{z})\right]}_{\text{reconstruction term}} - \underbrace{D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x}) \parallel p(\boldsymbol{z}))}_{\text{prior matching term}} \quad \text{(Definition of KL Divergence)}$$

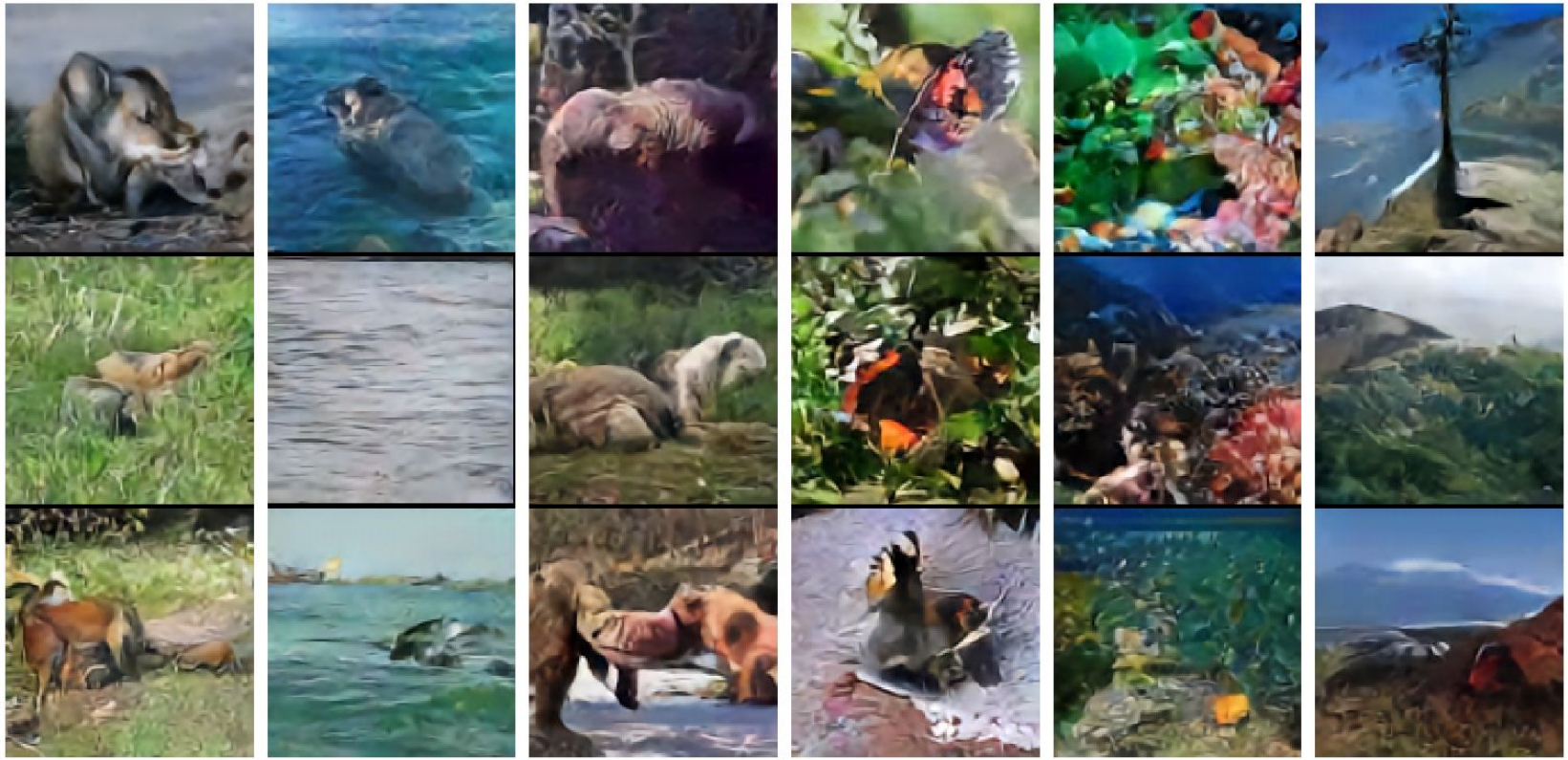
(We are **maximizing** this lower bound.)

If we maximize $p_\theta(\mathbf{x}|\mathbf{z})$ and minimize the D_{KL} we get close to $P(\mathbf{x})$.



[Calvin Luo https://arxiv.org/abs/2208.11970]

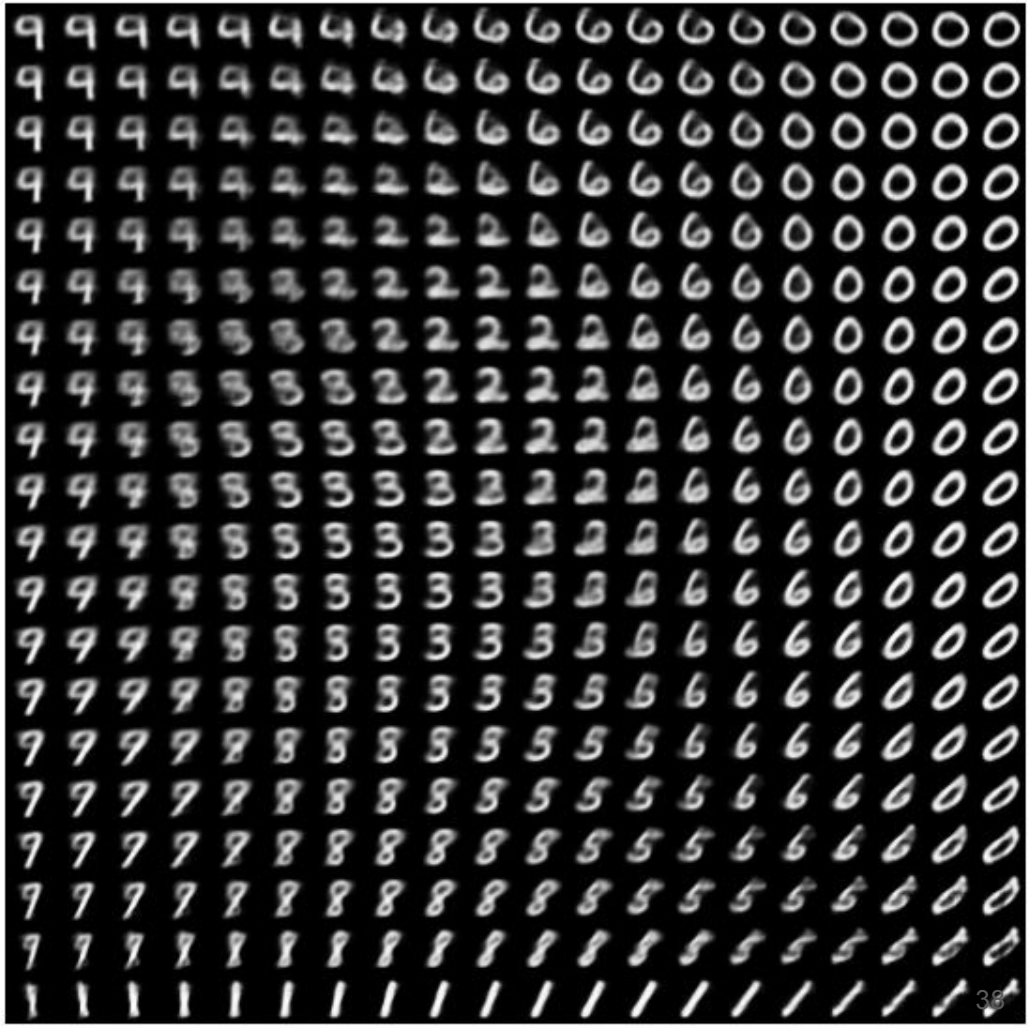
Examples of VAE generated images



a much nicer space…

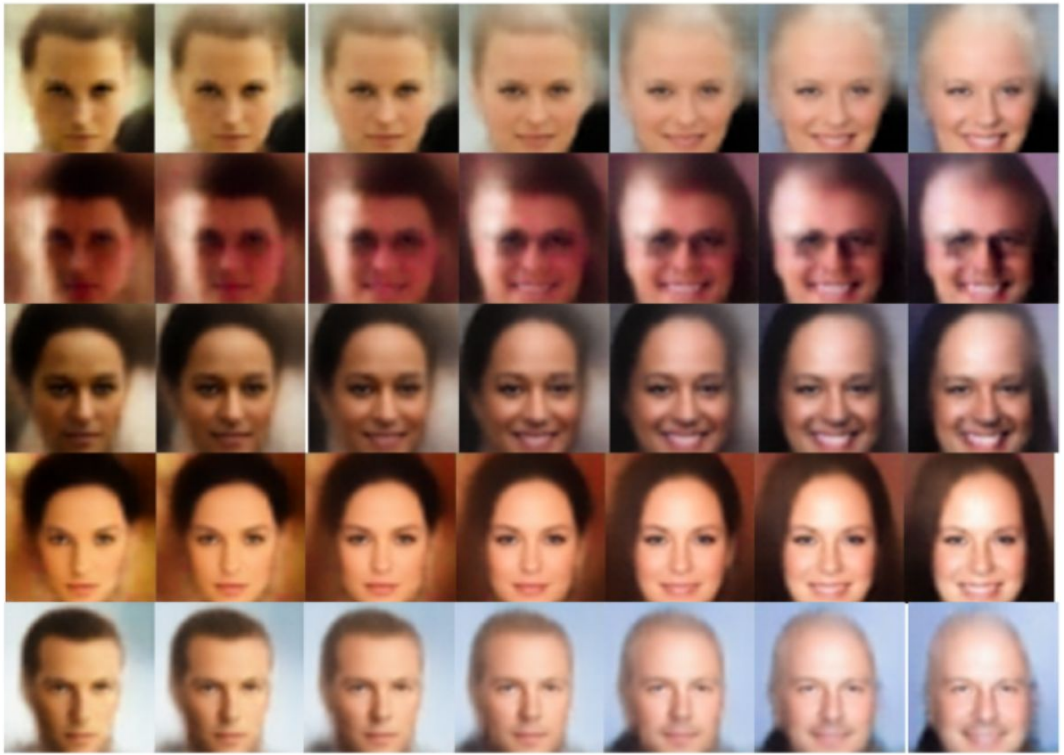
can smoothly interpolate digits in a meaningful, digit-y kind of way

DEMO



a much nicer space

dimensions in latent space correspond to meaningful concepts, like sentiment and orientation

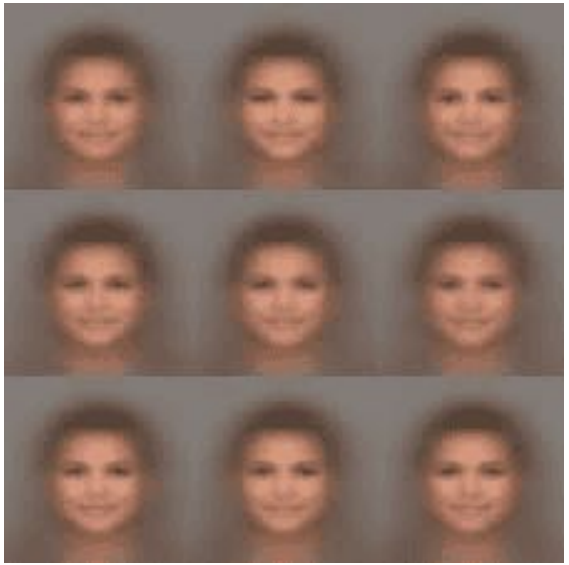


The Biggest Drawback of VAEs

- Out of the box, generated images can be blurry.

Question: Why? How do GANs fix this problem?

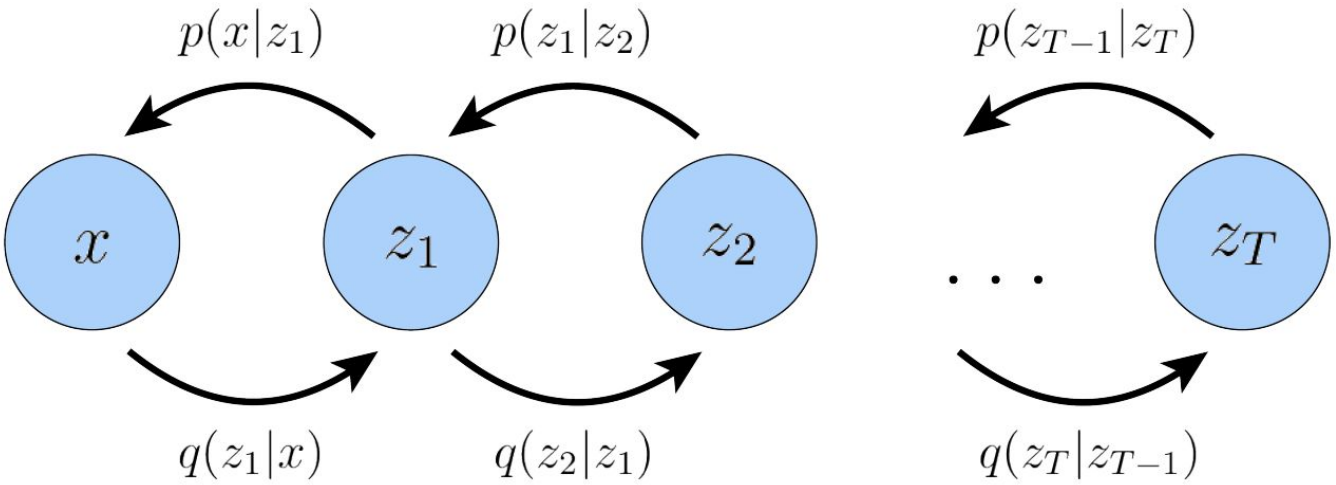
VAE v. GAN



https://borisburkov.net/2022-12-31-1/

Hierarchical VAEs

The generative process is modeled as a Markov chain, where each latent z_t is generated only from the previous latent z_{t+1}



Summary

- Generative Image models learn a mapping from the **Standard Normal Gaussian** to the **Image Manifold**
  - GANs learn this through a **discriminator**.
  - VAEs learn it through **variational autoencoders**
- AutoEncoders learn to **compress** and **reconstruct** data
- VAEs make these AutoEncoders **probabilistic**
  - Minimize the **reconstruction loss**
  - Latent space is sampled from Gaussian distributions
  - Sampling is made differentiable with the **Reparameterization Trick**
  - Deviations from the Prior (Standard Normal Gaussian) is penalized by **KL divergence**
- The ELBO is a **lower bound** of P(X)
  - Maximizing the ELBO, and minimizing the KL divergence makes P(x|z) close to P(x)